KAIZEN MATHS

Lesson starter

Loci and Constructions

Loci

Name _____

SA1

Draw the locus of points at the stated distance from the point.



Draw the locus of points 2 cm from point P.

Show your working

KAIZEN MATHS

Lesson starter

Loci and Constructions

Loci

Name _____

SB1

Draw the locus of points at the stated distance from the line.



Draw the locus of points 4 cm from line l.

Show your working

KAIZEN MATHS

Lesson starter

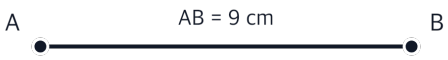
Loci and Constructions

Loci

Name _____

SC1

Draw the locus of points equidistant from the two fixed points.



Draw the locus of points equidistant from A and B.

GEO-F01A-L2 · SC

Show your working

KAIZEN MATHS

Lesson starter

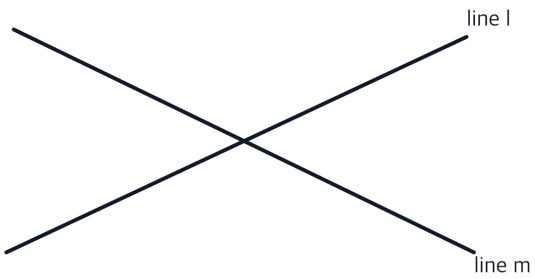
Loci and Constructions

Loci

Name _____

SD1

Draw the locus of points equidistant from the two intersecting lines.



Draw the locus of points equidistant from line l and line m.

GEO-F01A-L2 · SD

Show your working

KAIZEN MATHS

Exit ticket

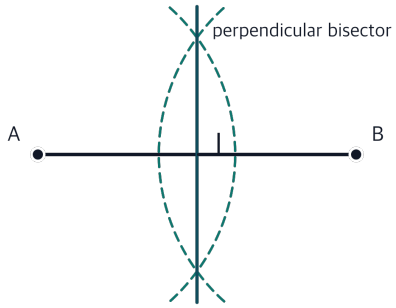
Loci and Constructions

Loci

Name _____

EA1

Describe the complete locus shown in the diagram.



Describe the locus shown.

Show your working

KAIZEN MATHS

Exit ticket

Loci and Constructions

Loci

Name _____

EB1

Draw the locus of points at the stated distance from the point.



Draw the locus of points 5 cm from point P.

Show your working

KAIZEN MATHS

Exit ticket

Loci and Constructions

Loci

Name _____

EC1

Draw the locus of points at the stated distance from the line.



Draw the locus of points 3 cm from line l.

Show your working

KAIZEN MATHS

Exit ticket

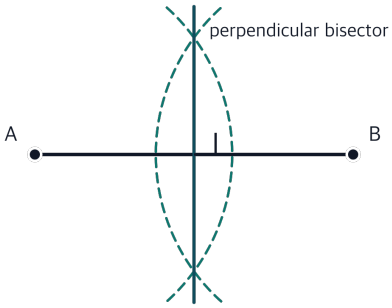
Loci and Constructions

Loci

Name _____

EA1

Describe the complete locus shown in the diagram.



Describe the locus shown.

Show your working

Teacher answers & working

Loci and Constructions · Loci

GEO-F01A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Draw the locus of points at the stated distance from the point.



Draw the locus of points 2 cm from point P.

Answer:

A circle with centre P and radius 2 cm.

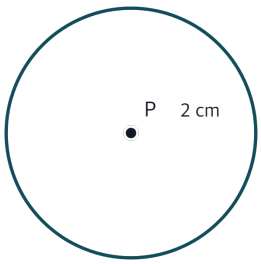
Working:

A fixed distance from a single point creates a circle.

The fixed point is the centre of the circle.

Set the compass width to 2 cm.

Place the compass point on P and draw a complete circle.



The locus is the circle with centre P and radius 2 cm.

Teacher answers & working

Loci and Constructions · Loci

GEO-F01A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Draw the locus of points at the stated distance from the line.



Draw the locus of points 4 cm from line l.

Answer:

Two lines parallel to line l, each 4 cm from l.

Working:

A fixed distance from a straight line creates two parallel lines.

Measure 4 cm perpendicular to line l on one side and draw a parallel line.

Measure 4 cm perpendicular to line l on the other side and draw a second parallel line.



The locus is the pair of parallel lines 4 cm from line l.

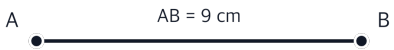
Teacher answers & working

Loci and Constructions · Loci

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SC1

Draw the locus of points equidistant from the two fixed points.



Draw the locus of points equidistant from A and B.

Answer:

The perpendicular bisector of AB.

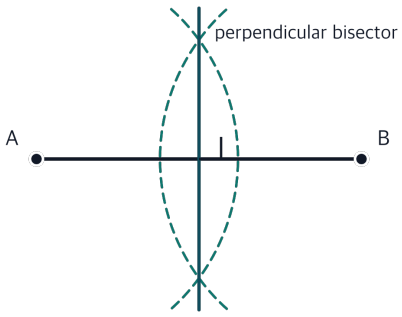
Working:

Points equidistant from A and B lie on the perpendicular bisector of AB.

Set the compass radius to more than half AB.

Draw equal arcs from A and B above and below AB.

Join the two arc intersection points.



Every point on this line is the same distance from A as it is from B.

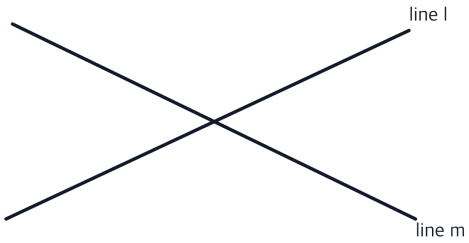
Teacher answers & working

Loci and Constructions · Loci

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SD1

Draw the locus of points equidistant from the two intersecting lines.



Draw the locus of points equidistant from line l and line m.

Answer:

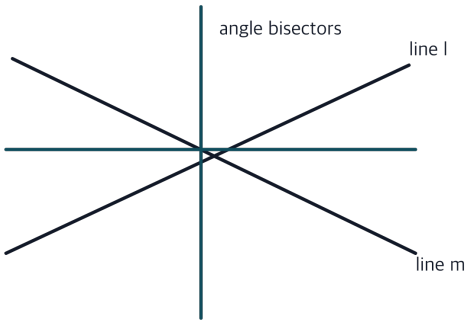
The angle bisectors of the angles between the two lines.

Working:

Points equidistant from two intersecting lines lie on the angle bisectors.

Construct one angle bisector using equal arcs from the two arms of the angle.

Construct the opposite angle bisector in the same way.



The complete locus is the pair of angle bisectors.

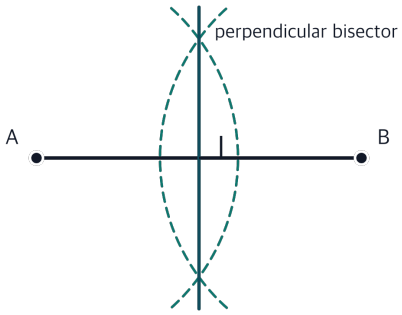
Teacher answers & working

Loci and Constructions · Loci

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EA1

Describe the complete locus shown in the diagram.



Describe the locus shown.

Answer:

The set of points equidistant from A and B.

Working:

The line crosses AB at 90 degrees and cuts AB into two equal parts.

This is the perpendicular bisector of AB.

Every point on a perpendicular bisector is equidistant from the two endpoints.

Therefore the locus is the set of points equidistant from A and B.

Teacher answers & working

Loci and Constructions · Loci

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EB1

Draw the locus of points at the stated distance from the point.



Draw the locus of points 5 cm from point P.

Answer:

A circle with centre P and radius 5 cm.

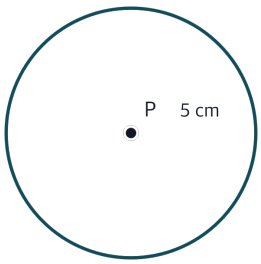
Working:

A fixed distance from a single point creates a circle.

The fixed point is the centre of the circle.

Set the compass width to 5 cm.

Place the compass point on P and draw a complete circle.



The locus is the circle with centre P and radius 5 cm.

Teacher answers & working

Loci and Constructions · Loci

GEO-F01A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Draw the locus of points at the stated distance from the line.



Draw the locus of points 3 cm from line l.

Answer:

Two lines parallel to line l, each 3 cm from l.

Working:

A fixed distance from a straight line creates two parallel lines.

Measure 3 cm perpendicular to line l on one side and draw a parallel line.

Measure 3 cm perpendicular to line l on the other side and draw a second parallel line.



The locus is the pair of parallel lines 3 cm from line l.